北京交通大学

硕 士 专 业 学 位 论 文

面向复杂真实场景的停车位检测技术研究

Research on Parking Slot Detection Techniques for Complex Real-World Scenarios

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北京交通大学

2026 年 5 月

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Research on Parking Slot Detection Techniques for Complex Real-World Scenarios

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致谢

落笔终章。随着硕士毕业论文的完工，这场持续了二十多年的学习生涯，应该大概率要告一段落了。首先要感谢一下，陪着我一路走来的老师们、朋友们，以及在背后默默付出的父母，以及所有在学习路上对我给予过帮助的人，正是有你们的支持，让我一路走到现在。当然，也应该感谢一下自己，这么多年，辛苦了。但是感谢归感谢，站在这个人生的路口，更重要的，是留下一点儿自己的感悟，对人生，对这个世界。

回顾这二十多年的光阴，我有无数次的迷茫、焦虑，也有无数次的幸福、喜悦。在那个 2016 年的夏天，我面临了人生第一次决定命运的考试，中考。考试前的那天晚上，是我第一次失眠。失眠只有 0 次和无数次，所以，我认为是那次考试开始，剥夺了我的好睡眠。但总归，结果还不错，有失就有得嘛。所以，对于我现在所获的的小小的学业路上的成就，也是用了很大的代价换来的，我不好评判这是否值得。失去的东西就让它随风而去吧，我想思考一下，我得到了什么。

那些所谓的成绩、荣誉，我觉得，以及没什么意义了，也就是当时能让自己暗自窃喜。我觉得，这二十年来最重要的，是给了我一个自己的视角去看待这个世界。说的简单一些吧，是一个让我了解自己的过程。我对于人生的意义，略微窥探到了一丝一毫。这里我想引用《明朝那些事儿》中作者的最后一句话：

成功只有一个——按照自己的方式，去度过人生。

摘要

随着自动泊车与智能驾驶技术的快速演进，停车位检测作为环境感知系统的核心环节，其检测精度与鲁棒性直接决定了自动泊车系统的安全性与可靠性。与传统目标检测任务不同，停车位检测具有强结构约束特性，同时面临标注成本高昂、场景分布呈现长尾效应等挑战，导致现有方法在复杂真实环境下仍存在数据规模受限、模型泛化能力不足以及检测结果结构不一致等问题。针对上述挑战，本文围绕停车位检测任务，从数据构建与学习范式、端到端结构建模以及多帧时序信息融合三个层面开展系统性研究:

（1）构建大规模复杂场景停车位数据集并提出半监督学习范式。针对现有数据集场景覆盖有限、人工标注成本高昂的瓶颈，本文构建了包含 35379 张图像的大规模真实场景停车位检测数据集 CRPS-D，具备场景类型丰富、停车位密度高以及异常车位比例大等特点。在此基础上，结合停车位关键点分布稀疏且结构约束显著的任务特性，提出了一种基于教师–学生架构的半监督学习框架（SS-PSD）。SS-PSD 通过置信度引导掩码一致性与自适应特征扰动机制，有效挖掘了无标注数据中的空间结构先验。实验结果表明，在仅使用 1/12 标注训练数据的情况下，模型在 ps2.0 数据集上的检测平均精度（AP）达到 90.47%，在 CRPS-D 数据集上的 AP 达到 79.75%，在相同标注比例设置下显著优于对应的全监督基线模型。

（2）设计单帧结构感知的端到端停车位检测方法。针对现有方法依赖人工设计后处理规则、难以保证结构一致性的局限，本文提出了一种结构感知的端到端检测框架。该方法在网络内部显式建模几何拓扑约束，实现了停车位关键点定位与结构关系的联合优化，从而摒弃了传统启发式匹配流程。实验结果表明，该方法在 ps2.0 数据集上取得了 99.37% 的查准率、99.47% 的查全率以及 99.47% 的 AP；在更具挑战性的 CRPS-D 数据集上，分别取得了 92.21% 的查准率、85.54% 的查全率以及 83.58% 的 AP，整体性能显著优于现有端到端方法。

（3）提出多帧时序特征融合的停车位检测增强方法。针对单帧检测易受瞬时遮挡和环境噪声干扰的问题，本文进一步将静态空间建模扩展至时空联合建模，提出了一种多帧时序特征融合方法。该方法通过挖掘连续视频序列中的时序相关性，对结构感知特征进行递归建模与时序平滑，有效提升了检测结果在时间维度上的一致性。在自建时序数据集 PSTD 上的实验结果表明，该方法取得了 92.53% 的查准率、92.33% 的查全率以及 91.97% 的 AP。同时，该模型在车载嵌入式平台上实现了约 30 FPS 的稳定推理速度，满足自动泊车系统的实时应用需求。

关键词：停车位检测；数据集；半监督学习；关键点检测；结构感知；时序建模

ABSTRACT

With the rapid evolution of automatic parking and intelligent driving technologies, parking slot detection serves as a core component of the environmental perception system. Its detection accuracy and robustness directly determine the safety and reliability of automatic parking systems. Unlike traditional object detection tasks, parking slot detection exhibits strong structural constraints while facing challenges such as high annotation costs and long-tailed scene distribution. These issues lead to limitations of existing methods in complex real-world scenarios, including insufficient data scale, poor model generalization, and inconsistent detection structure. To address the above challenges, this paper conducts systematic research on parking slot detection from three perspectives: data construction and learning paradigm, end-to-end structural modeling, and multi-frame temporal information fusion:

(1) Construction of a large-scale complex-scene parking slot dataset and proposal of a semi-supervised learning paradigm. To tackle the bottlenecks of limited scene coverage and high manual annotation costs in existing datasets, this paper builds CRPS-D, a large-scale real-world parking slot detection dataset containing 35,379 images. CRPS-D features diverse scene types, high parking slot density, and a large proportion of abnormal slots. On this basis, combined with the task characteristics of sparse key point distribution and significant structural constraints in parking slots, a Teacher-Student based Semi-Supervised Parking Slot Detection framework (SS-PSD) is proposed. By leveraging confidence-guided mask consistency and adaptive feature perturbation, SS-PSD effectively excavates spatial structural priors from unlabeled data. Experimental results show that using only 1/12 annotated training data, the model achieves an Average Precision (AP) of 90.47% on the ps2.0 dataset and 79.75% on the CRPS-D dataset, significantly outperforming the corresponding fully supervised baseline under the same annotation ratio.

(2) Design of an end-to-end parking slot detection method with single-frame structure awareness. Aiming at the limitation that existing methods rely on manually designed post-processing rules and fail to guarantee structural consistency, this paper proposes a structure-aware end-to-end detection framework. The method explicitly models geometric topological constraints within the network, enabling joint optimization of parking slot key point localization and structural relationships, thus eliminating traditional heuristic matching pipelines. Experimental results demonstrate that the proposed method achieves

a precision of 99.37%, recall of 99.47%, and AP of 99.47% on the ps2.0 dataset. On the more challenging CRPS-D dataset, it obtains a precision of 92.21%, recall of 85.54%, and AP of 83.58%, substantially surpassing state-of-the-art end-to-end methods in overall performance.

(3) Proposal of a multi-frame temporal feature fusion enhancement method for parking slot detection. To mitigate the susceptibility of single-frame detection to instantaneous occlusion and environmental noise, this paper further extends static spatial modeling to joint spatio-temporal modeling and proposes a multi-frame temporal feature fusion method. By exploring temporal correlation in consecutive video sequences, the method performs recursive modeling and temporal smoothing on structure-aware features, effectively improving the temporal consistency of detection results. Experimental results on the self-built temporal parking slot dataset PSTD show that the method achieves a precision of 92.53%, recall of 92.33%, and AP of 91.97%. Meanwhile, the model realizes stable inference at approximately 30 FPS on vehicle embedded platforms, meeting the real-time application requirements of automatic parking systems.

KEYWORDS: Parking Slot Detection; Dataset; Semi-supervised Learning; Keypoint Detection; Structure-aware Modeling; Temporal Modeling

目录

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二、发表论文

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三、参与科研项目

[1] 鱼眼图像处理的关键技术研究，国家自然科学基金

[2] 智能驾驶场景下的自动标注关键技术研究，毫末智行科技有限公司

四、参与竞赛项目

[1] 第五届“华为杯”中国研究生人工智能创新大赛（2023）国家级二等奖

[2] 第十九届“挑战杯”全国大学生课外学术科技作品竞赛（2024）国家级二等奖

五、获奖情况

(1) 一等学业奖学金，北京交通大学，2024

(2) 硕士研究生国家奖学金，中华人民共和国教育部，2025

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